Chapter 3: Model Questions	
Q1	Generate the voltage values corresponding to: $V = 230 \sin(\theta)$ for **9 equally spaced angles between 0° and 360°**. Use linspace() to generate the angle array.
Q2.	Write a NumPy program to generate 9 equally spaced values of x in the interval [0-pi]. Use array slicing to select alternate elements and compute both sin(x), sinh(x), cos(x) and cosh(x)n for the selected elements. Display all the results with suitable labels.
Q3.	The temperature recorded at 8 different locations over 7 days is stored in a NumPy array of shape 8×7. Write a NumPy program to: 1. Generate or create the temperature dataset. 2. Calculate the **average temperature for each location**. 3. Find the **maximum and minimum temperature** recorded at each location. 4. Find the location having the **highest average temperature**. 5. Count the number of temperature readings above 35 degree Celsius.
Q4.	A grayscale image is represented as a 6×6 NumPy array, where each element represents a pixel intensity between 0 and 255. Write a program to: 1. Create a 6×6 grayscale image matrix. 2. Convert the image into a **binary image** using a threshold of 128. Pixels ≥ 128 should become 1 and the remaining pixels should become 0. 3. Find the **maximum, minimum, and average pixel intensity**. 4. Extract the central 4×4 portion of the image using NumPy slicing. 5. Increase the brightness of the image by 30, ensuring that no pixel value exceeds 255.
Q5.	Consider the following system of linear equations: $2x + y + z = 10$ $x + 3y + 2z = 15$ $x + y + 4z = 20$ Using **NumPy**, write a program to: 1. Represent the coefficient matrix A and constant matrix B. 2. Calculate the **determinant of A**. 3. Calculate the **inverse of A**, if it exists. 4. Solve the system of equations using numpy.linalg.solve(). 5. Verify the obtained solution by calculating AX.

Q6.

Given the following vectors

$$A = (2, 1, 3)$$
$$B = (1, -1, 2)$$
$$C = (3, 2, -1)$$

Using NumPy:

1. Calculate A . B.
2. Calculate A x B.
3. Calculate A . (B x C).
4. Calculate (A x B) . C.
5. Verify the identity: A . (B x C) = (A x B) . C
6. State the geometrical meaning of the final result.

Q7.

. In a mechanical system, a force F = (10, 20, 30) N is applied at a point whose position vector from the origin is r = (2, 3, 1) m.

1. Calculate the magnitude of the torque.
2. If the displacement vector is d = (3, 1, 2) m, calculate the work done using:

Work = F . d

5. Calculate the angle between F and d.
6. Explain the physical meaning of the dot product and cross product used in this problem.

Q8.

For two vectors A and B, calculate and compare their dot product, cross product, and dyadic product. Explain why the dimensions and physical meanings of the results are different.

Q9.

A drone is programmed to fly with an intended velocity

$$\vec{v}_d = 12\hat{i} + 5\hat{j}\ m/s.$$

However, wind produces a velocity

$$\vec{v}_w = 3\hat{i} - 2\hat{j}\ m/s.$$

The actual velocity of the drone is

$$\vec{v}_a = \vec{v}_d + \vec{v}_w.$$

Write a Python/NumPy program to:

1. Calculate the actual velocity vector.
2. Calculate its magnitude.
3. Find the unit vector representing the actual direction of motion.
4. Calculate the angle between the intended velocity and actual velocity.

Q10.	Write a Python program using **NumPy** to generate and analyze the marks of 50 **students in three different subjects**. The marks should be generated from a **normal distribution** with: • Mean =8 • Standard deviation = 5 The program should perform the following tasks: 1. **Generate the student marks** for all three subjects using NumPy's normal-distribution random number generator. 2. Store the generated marks in a **2-dimensional NumPy array**, where rows represent students and columns represent subjects. 3. Determine the **maximum and minimum marks** obtained in the complete dataset and identify their **student and subject locations**. 4. Calculate the **variance** and **standard deviation** of the generated marks. 5. Calculate the **mean and median** of the generated marks. 6. Display the results with appropriate labels.
Q.11	A force $$\vec{F} = 50\hat{i} + 30\hat{j} - 20\hat{k}\ N$$ acts on an object. The object undergoes a displacement $$\vec{d} = 4\hat{i} + 6\hat{j} + 3\hat{k}\ m.$$ Using NumPy: 1. Calculate the magnitude of the force. 2. Calculate the **work done** by the force using the dot product: $$W = \vec{F} \cdot \vec{d}.$$ 3. Determine whether the force does **positive or negative work**. 4. Calculate the angle between the force and displacement vectors.